PORTFOLIO WALKTHROUGH

Panic Attack Dataset Analysis

Descriptive Analytics using Snowflake SQL and Power BI

Sandip Mandal · 1,200 records · 21 columns

ANALYTICS STACK

Snowflake

SQL profiling and descriptive analysis

Power BI

Interactive dashboard reporting

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PROJECT QUESTION

What the project set out to answer

Use a sample dataset to profile the records, validate the fields, compare descriptive patterns, and turn the analysis into an interactive dashboard.



Profile and validate

Check completeness, uniqueness, and field consistency.



Calculate descriptives

Summarize duration, score, frequency, and records.



Compare patterns

Review triggers, demographics, sleep, history, medication, and therapy.



Build the dashboard

Translate analysis into an interactive Power BI experience.

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WORKFLOW

From CSV to four-page dashboard

Dataset: panic_attack_dataset.csv · 1,200 rows · 21 columns

Pipeline

1. CSV
2. Snowflake table
3. Snowflake SQL Worksheet
4. Power BI Import Mode
5. Four-page dashboard

Dataset fields

Age, Gender, Panic_Attack_Frequency, Duration_Minutes, Trigger, Heart_Rate, symptoms, Medical_History, Medication, Therapy, Sleep_Hours, Panic_Score, and lifestyle fields.

Why Snowflake: SQL supports repeatable profiling and transparent transformations on the sample records. **Why Import Mode:** Power BI Import Mode supports responsive slicing and fast dashboard interaction.

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DATA QUALITY

The source is complete on IDs, but not on medical history

1,200
Total records

1,200
Unique IDs

122
Blank Medical_History values

Validation check	Finding
No duplicate IDs observed	IDs are complete and unique in the uploaded source CSV.
Missing medical history	**Not recorded**, unless source documentation defines otherwise.
Consistent groups	Age: 18–25, 26–35, 36–45, 46–55, 56–64; duration: Short 5–14, Medium 15–24, Long 25–34, Very Long 35–44 minutes.

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KPI ANCHOR · RECORDED DURATION

24.39

minutes average recorded duration

Median: 24 minutes · Minimum: 5 · Maximum: 44

The typical recorded attack lasts about 24 minutes

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5.57

Average panic score

4.41

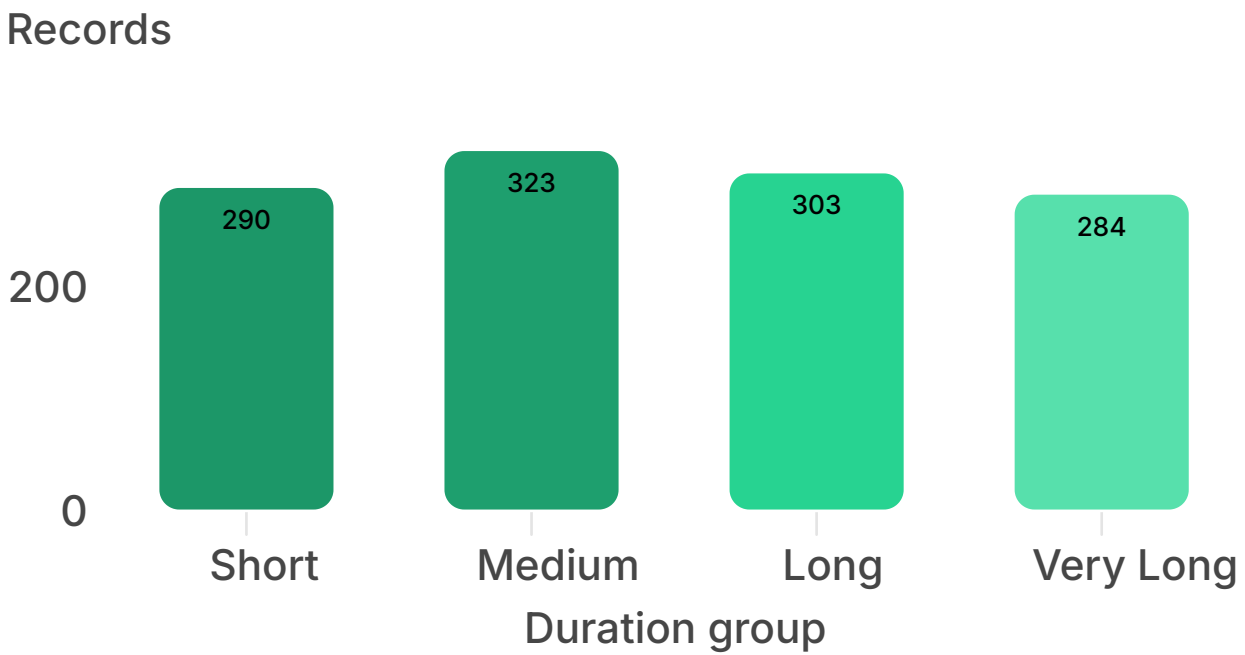
Average attack frequency

Sample standard deviation: approximately 11.40 minutes. Time period for frequency was not documented.

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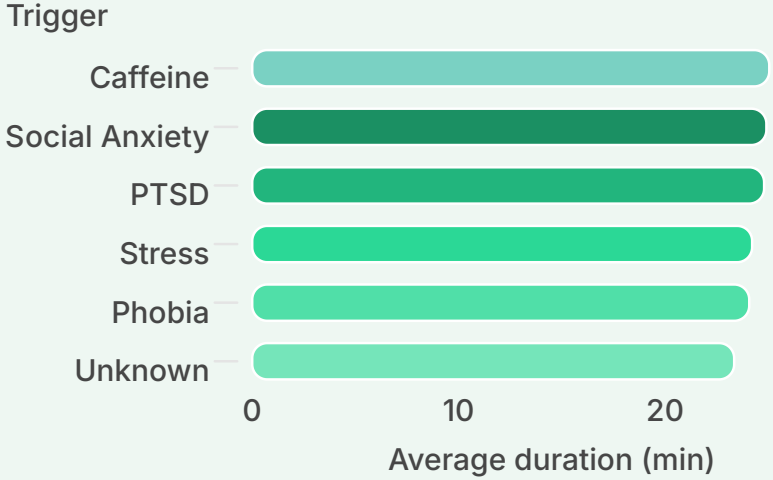
DURATION DISTRIBUTION

Most records sit in the middle of the duration range



Medium and Long records together: 626 records, or 52.17% of the sample.

Trigger averages stay close together



Differences are descriptive and do not establish causation.

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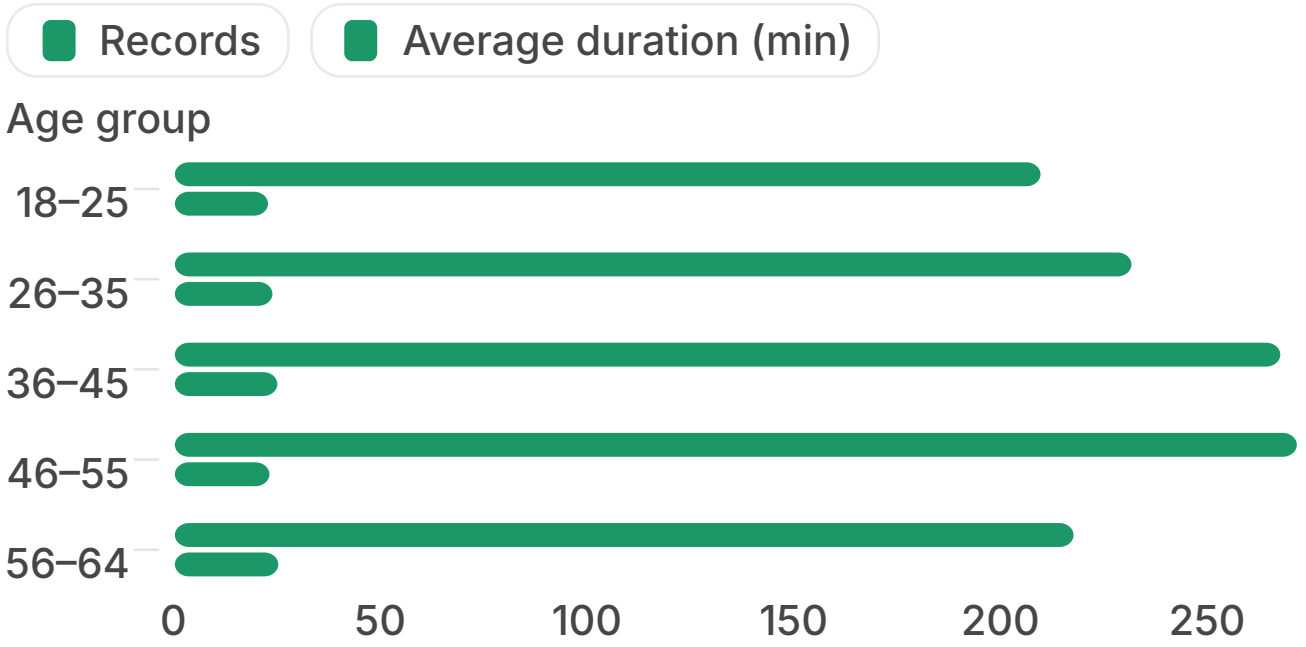
DEMOGRAPHIC COMPARISON

Age and gender differences are descriptive, not diagnostic

Gender snapshot

Gender	Records	Avg. duration
Female	549	24.86 min
Male	537	23.97 min
Non-binary	114	24.16 min

Average score: Female 5.63 · Male 5.53 · Non-binary 5.43



46–55 has the highest record count; 56–64 has the highest average recorded duration.

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Sleep and medical history need careful context

The 122 blank values are missing source values, not a confirmed medical category.

Sleep groups

Sleep group	Records	Avg. score	Avg. duration
Below 5 Hours	217	5.29	24.81 min
5–7 Hours	505	5.65	24.15 min
7+ Hours	478	5.61	24.46 min

Recorded medical history

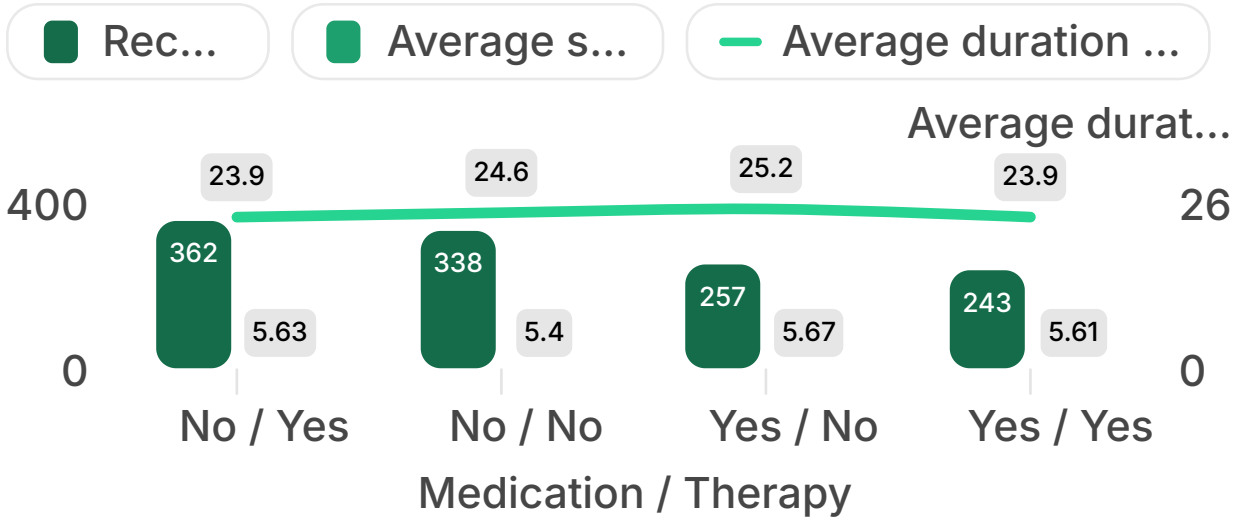
Recorded value	Records
Anxiety	492
Depression	349
PTSD	237
Blank / not recorded	122

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OBSERVATIONAL COMPARISON

Medication and therapy groups show observational differences

The groups differ descriptively, but the dataset cannot measure treatment effectiveness.



Sample-record differences are descriptive; the dataset cannot establish causation or treatment effectiveness.

Dashboard pages: Panic Attack Overview; Attack Duration & Trigger Analysis; Demographic Analysis; Lifestyle & Medical History Analysis.

Descriptive only

Reserve a clean area for actual Power BI screenshots if added later.

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CLOSING READOUT

The project demonstrates disciplined descriptive analytics

Profile the data. Make the comparisons visible. Keep the claims proportional to the source.

24.39 minutes

Average recorded duration

52.17%

Records between 15 and 34 minutes

46–55

Largest age group by record count; 56–64 has the highest average recorded duration

25.04 minutes

Highest average recorded duration among trigger groups: Caffeine

Limits: sample dataset; undocumented source and sampling method; 122 blank Medical_History values; field definitions and units needing confirmation; no causal or medical conclusions; no date field for time trends.

Conclusion: the project demonstrates data profiling, Snowflake SQL analysis, descriptive comparisons, and interactive Power BI reporting.

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